Face Detection & Recognition on DE2-115 — User Manual

Quartus project: `designfile_testing/vj.qpf` (revision `top`), all logic in Verilog (`.v`). Board: Terasic DE2-115 (Cyclone IV EP4CE115F29C7). Camera: OV7670 on the 40-pin GPIO header.

The system shows live camera video, finds faces (Viola-Jones), draws a green box around each face and labels it with the name of the enrolled person it matches and a match score — or `UNKNOWN`. Up to **4 people** can be enrolled; they are stored in the on-board flash and survive power-off. The design itself is stored in the EPCS boot memory, so the board starts the system on its own at power-on (no PC needed).

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1. Hardware setup

Item	Setting
SW19 (RUN/PROG), left edge near the LCD	**RUN** (otherwise the board does not boot the design)
Power	12 V adapter, red power button
Monitor	VGA connector (blue, 15-pin); select **VGA input** on the monitor
USB-Blaster	Only needed to reprogram the board (USB port marked *USB-Blaster*)

Camera wiring (OV7670 → GPIO header)

The FPGA pins below are what the design uses (`top.qsf`). **D7 is on GPIO[8] (header pin 9), not GPIO[7]** — header pin 8 on this board does not make reliable contact.

Camera pin	FPGA pin	GPIO	Header pin
D0	AB22	GPIO[0]	1
D1	AC15	GPIO[1]	2
D2	AB21	GPIO[2]	3
D3	Y17	GPIO[3]	4
D4	AC21	GPIO[4]	5
D5	Y16	GPIO[5]	6
D6	AD21	GPIO[6]	7
D7	**AD15**	**GPIO[8]**	**9**
PCLK	AC19	GPIO[10]	13
XCLK	AF16	GPIO[11]	14
PWDN	AD19	GPIO[12]	15
RESET	AF15	GPIO[13]	16
SIOC (SCL)	AF24	GPIO[14]	17
SIOD (SDA)	AE21	GPIO[15]	18
VSYNC	AF25	GPIO[16]	19
HREF	AC22	GPIO[17]	20
3.3 V / GND	—	—	29 / 30

Use **LEDG0–LEDG7** (section 4) to check the data wires: all 8 must be lit with a normal scene in front of the camera.

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2. Power-on

1. Turn the board on. Within about a second:
   - **LEDR0** (PLL locked) and **LEDR1** (camera configured) light,
   - **LEDR5** flashes briefly while the enrolled people are read from flash,
   - live video appears on the monitor.
2. With people in view, **LEDR2** starts toggling (one change per detection pass) and green boxes appear.

Screen layout

- **Left, 320×240:** live camera video. Each detected face gets a green box and a label `NAME NN%` on a black strip above it (below it if the box touches the top edge).
- **Right:** enrollment gallery — 4 slots, each with the stored photo (or a dark square), the name and `ENROLLED` / `EMPTY`. The slot selected for enrollment has a grey border, **yellow** while enrollment is waiting.

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3. Switches and keys

Switches are the slide switches along the bottom edge; **up = 1**. Unlisted switches have no function.

Control	Function
SW17	Reset. Keep **down**. Up holds the system in reset (no detection).
SW16	Resend camera registers. Hold **up for ~1 s** then down — use if colours look wrong after the camera was re-plugged.
SW15	Capture mode. **Down = continuous detection** (normal). Up = detect one frame each time KEY0 is pressed.
SW14	Test image. Up = freeze the screen and the detector on the built-in face-grid test image (camera ignored). Normal: down.
SW13	Overlay. Up = show boxes and labels. Down = clean live video.
SW11–SW10	Enrollment slot (see table below).
SW2–SW0	Minimum face-group size (binary 0–7). A face is shown only if more raw detections than this agree. Recommended **3** (SW0 + SW1 up). Raise if stray boxes appear, lower if real faces are missed.
SW3–SW9, SW12	Not used.
KEY0	Capture one frame (only when SW15 is up).
KEY1	Enroll the person in front of the camera into the selected slot (section 5).
KEY2	Not used.
KEY3	**Forget everyone**: clears all 4 slots and erases them from flash.

Enrollment slot (names are set in `person_names.v`, section 7):

SW11	SW10	Slot	Name (current)
down	down	0	ANNET MARIA
down	up	1	ANITA PAUL

SW11	SW10	Slot	Name (current)
up	down	2	SHAMEER
up	up	3	NAIDU SIR

The recognition threshold is fixed at **75 %** — there is nothing to set.

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4. LEDs

Red LEDs (LEDR)

LED	Meaning
LEDR0	PLL locked (clocks running). Always on in normal operation.
LEDR1	Camera configuration finished. Always on in normal operation.
LEDR2	Detection pass completed — **toggles** after every pass (about every 1–2 s). Steady = detection is not running (check SW17/SW15).
LEDR3	The last pass found at least one raw face detection.
LEDR4	Enrollment waiting / in progress (KEY1 pressed). Goes off after 4 views are stored.
LEDR5	Flash busy: reading at power-on, saving after an enrollment (1–2 s), or erasing (KEY3).
LEDR6–LEDR9	Main state machine state (diagnostic, binary, LEDR6 = bit 0). Flickers in normal operation.
LEDR10–LEDR13	Recognizer state (diagnostic, binary, LEDR10 = bit 0). Flickers when faces are present.
LEDR14–LEDR17	Not used.

Green LEDs (LEDG)

LED	Meaning
LEDG0–LEDG7	Camera data line D0–D7 is toggling. All 8 should be lit with a normal scene. A **dark or blinking** LED means that data wire is loose or on the wrong header pin (wrong colours, cyan highlights, speckle).
LEDG8	Not used.

Diagnostic state codes (only needed for troubleshooting)

LEDR6–9 = main state machine (value = LEDR9·8 + LEDR8·4 + LEDR7·2 + LEDR6):

Value	State	If it stays here
0	RESET	SW17 is up
1	CAPTURE_START	SW15 is up and KEY0 not pressed, or no camera frames (check PCLK/VSYNC wires)
2	CAPTURE	camera frames not arriving
3–8	detection (integral image / cascade / next scale)	normal, changes quickly
9–10	FACEBOX_START / FACEBOX	normal, changes quickly
11	POST	waiting for flash (LEDR5 on)
12	ENROLL	recognizer running (LEDR10–13 flicker)

LEDR10–13 = recognizer (value = LEDR13·8 + LEDR12·4 + LEDR11·2 + LEDR10): 0 idle, 1 next face, 2 crop, 3–6 LBP histogram, 7 store enrollment view, 8 compare, 9–10 search positions, 11–13 score, 14 done.

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5. Enrolling people

Enroll each person once; they stay stored across power-off.

1. Set **SW13 up** (overlay), **SW15 down**, **SW17 down**.
2. Select the slot with **SW11–SW10** (section 3). The slot gets a grey border in the gallery.
3. The person stands **alone** in front of the camera, at the distance and **lighting** that will be used later, until a green box appears.
4. Press **KEY1** once. **LEDR4** lights and the border turns **yellow**.
5. **While LEDR4 is on (about 4 detection passes, a few seconds), the person slowly changes pose:** looking straight → head a little left → a little right → chin a little up or down. Each pass stores one of 4 separate views. Keep the movements small so the face stays detected.
6. LEDR4 goes off, the photo appears with `ENROLLED`, and **LEDR5** lights for 1–2 s while it is saved to flash.
7. Select the next slot and repeat.
8. **Re-enroll** a person: select their slot and press KEY1 again — the 4 new views replace the old ones.
9. **Forget everyone:** press **KEY3** (all slots `EMPTY`, flash erased).
10. If LEDR4 stays on, no face is being detected — move closer, improve lighting, check SW13/SW15/SW17.

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6. Recognition

- Every detected face (up to 4 at once) is compared with every enrolled person after each detection pass.
- The label shows the best matching name and its **match score**. A score of **75 % or more** shows the name; below 75 % the label shows `UNKNOWN` with the score.
- Typical scores: enrolled person **80–95 %**, strangers **55–75 %**. The score is a similarity value, not a probability.
- Measured on a 40-person face database with this exact algorithm: correct name for enrolled people about 95 %, about 85 % correct overall including rejecting strangers. On the board, results are best when enrollment and use happen in the **same lighting**, faces look roughly at the camera, and faces are at least ~50 pixels wide on the screen.

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7. Changing names, rebuilding, reprogramming

Change a name

Edit `designfile_testing/person_names.v` — each name is **exactly 12 characters** (pad with spaces), letters A–Z, digits, space and `- . _`:

```
localparam [95:0] NAME0 = "ANNET MARIA ";
```

Enrolled faces are not affected by a name change (they stay in flash, linked to the slot).

Rebuild and make it permanent (from a terminal)

```
cd "C:/Users/Bless/Downloads/designfile_testing (2)/designfile_testing"
```

```
D:/intelFPGA/quartus/bin64/quartus_sh.exe --flow compile vj -c top
```

Check in `output_files/top.sta.rpt` that every clock has positive setup slack, then write the boot memory and load it:

```
D:/intelFPGA/quartus/bin64/quartus_cpf.exe -c -d EPCS64 -s EP4CE115 output_files/top.sof output_files/top.jic
```

```
D:/intelFPGA/quartus/bin64/quartus_pgm.exe -c "USB-Blaster [USB-0]" -m JTAG -o "pvi;output_files/top.jic"
```

```
D:/intelFPGA/quartus/bin64/quartus_pgm.exe -c "USB-Blaster [USB-0]" -m JTAG -o "p;output_files/top.sof"
```

The last command starts the new design immediately; otherwise it starts at the next power-on. If a programming step fails with "Can't scan JTAG chain", run it again (the USB-Blaster connection occasionally drops). Enrolled people survive reprogramming — they are in the separate parallel flash.

The USB-Blaster driver that works on this PC is the re-signed one from `C:\intelFPGA\22.1std\qprogrammer\drivers\usb-blaster` (the Quartus 20.1 driver's certificate has expired).

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8. Troubleshooting

Symptom	Check
Monitor says "no signal"	Monitor input set to VGA; VGA cable seated; LEDR0 on.
Black screen, LEDs dark after power-on	SW19 on RUN; power-cycle fully (blue LED off for 5 s).
Colours wrong (cyan highlights, green/magenta, speckle)	LEDG0–7 all lit? Reset the wire of a dark/blinking LEDG. Toggle SW16 to resend camera settings.
No boxes	SW13 up; LEDR2 toggling? If not: SW17 and SW15 down. Try SW2–SW0 lower.
Many stray boxes	Raise SW2–SW0 (e.g. 4 or 5).
Everyone `UNKNOWN`	Enroll again in the current lighting (section 5), with the 4 pose changes.
Wrong name	Re-enroll that person and the person it confuses them with, same lighting as use.
LEDR4 stays on	No face detected during enrollment — see section 5.
Enrolled people missing after power-on	LEDR5 should flash at power-on; re-enroll once (old saves from earlier design versions are ignored).

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9. Design files (in `designfile_testing/`)

File	Role
`top.v`	Top level: data flow, clocks, main state machine, switches/LEDs
`ov7670_controller.v`, `ov7670_registers.v`, `i2c_sender.v`	Camera configuration over SCCB

File	Role
`ov7670_capture.v`	Camera capture, 320×240 colour + full-precision grey
`dp_ram.v`	Dual-clock RAM used for all frame buffers and tables
`vga.v`, `Address_Generator.v`, `RGB.v`	VGA timing and output
`ii_gen.v`, `pixel_offset.v`	Integral images for detection
`subwindow_top.v`, `subwindow.v`, `feature_calc.v`, `rect_calc.v`, `var_norm_calc.v`, `register_file*.v`, `ii_address_*.v`, `iix2_address_decoder.v`, `parallel_dataPath_mux.v`, `ram.v`	Viola-Jones cascade (16 subwindows in parallel); parameters in the `.txt` / `.mif` files
`faceBox.v`	Collects raw detections and replays them as boxes
`face_cluster.v`	Groups raw detections into faces (min group size SW2–SW0)
`face_crop.v`, `lbp_uniform.v`, `face_recognizer.v`	Face crop and LBP recognition (4 views per person, fixed 75 % threshold)
`flash_store.v`	Saves/loads enrolled people in the parallel flash
`box_overlay.v`, `enroll_gallery.v`, `font8x8.v`, `person_names.v`	On-screen boxes, labels, gallery, font, names
`cam_bit_monitor.v`	LEDG camera-wire monitor
`my_altpll.*`, `ii_buffer.*`, `iix2_buffer.*`, `faceBox_buff.*`, `altsqrt_varianceCalc.*` (`.v`, `.qip`, `.bsf`, `_inst.v`, `_bb.v`, `my_altpll.ppf`)	Quartus MegaWizard-generated PLL/RAM/square root — do not edit by hand, keep each set complete

Data and project files:

File	Role
`*_rect*.txt`, `weight*.txt`, `weakThresh.txt`, `weakStageNum.txt`, `strongThresh.txt`, `left_tree.txt`, `right_tree.txt`	Cascade parameters loaded into the ROMs (`ram.v`, `$readmemh`) — **required**
same names `.mif`	Original source of those parameters (the `.txt` files were generated from them)
`test_image_grey8.mif`, `test_image_rgb444.mif`	Built-in face-grid test image (SW14)
`vj.qpf`, `top.qsf`, `top.out.sdc`	Quartus project, settings/pin assignments, timing constraints
`output_files/top.sof`, `output_files/top.jic`	Ready-to-program files (JTAG / EPCS boot memory)

Every `.v` file starts with a header comment describing its function, clock domain and where it is used. Unused files from the original design were removed; a full copy of the project before that clean-up is in `designfile_testing_full_backup_before_cleanup.zip` next to the project folder.